

Why $Q = 2/3$: The Koide Relation as a Topological Ratio of D_{IV}^5

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One claim. One derivation chain. Four pages.

Strategy (cold reader, May 5)

This is the door-opener paper. Requirements: - One narrow claim ($Q = \text{rank}/N_c$) - Short (4 pages, PRL format) - Self-contained (no reference to BST machinery beyond what's needed) - Sympathetic audience EXISTS (Koide community is active, ~50 papers/year cite the relation) - Falsifiable (if $Q \neq 2/3$ to higher precision, or if the Z_3 structure is wrong)

Outline

Abstract (~100 words)

We show that the Koide sum rule $Q = (m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ follows from the rank and color dimension of the type-IV bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. The ratio $Q = \text{rank}/N_c = 2/3$ is topological: it cannot receive perturbative corrections because both integers are invariants of the compact dual Q^5 . The Koide angle θ_0 (with $\cos(\theta_0) = -19/28$) decomposes into BST integers: $19 = n_C^2 - C_2$ and $28 = T_g$ (the g -th triangular number, also the perfect number from Mersenne prime $2^3 - 1 = 7 = g$). We predict that any future precision measurement of Q will converge toward $2/3 = 0.666\dots$, never away from it.

1. Introduction (~0.5 pages)

- Koide (1981): empirical observation that $Q = 2/3$ to 0.001%
- 40+ years of attempts to explain: discrete symmetries (Z_3 , S_3), texture zeros, democratic mass matrices
- No consensus mechanism; all models introduce new parameters
- We propose: $Q = 2/3$ is a topological ratio of the unique rank-2 Hermitian symmetric space in 5 complex dimensions

2. The Geometry (~1 page)

- $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$: bounded symmetric domain, type IV, rank 2, $\dim_C = 5$
- Five integers: rank = 2, $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$
- Compact dual Q^5 : smooth quadric in P^6 , $\chi(Q^5) = C_2 = 6$
- Key property: rank and N_c are TOPOLOGICAL invariants of Q^5 (not parameters)
- The Z_3 symmetry: $CP^2 = SU(3)/[SU(2) \times U(1)]$ is the color sector of D_{IV}^5 . The $N_c = 3$ color charge generates the Z_3 cyclic symmetry that appears in all Koide-type models.

3. The Derivation (~1.5 pages)

Step 1: The three charged leptons occupy the three Z_3 -orbits of the color sector of D_{IV}^5 . They are NOT colored (no strong interaction), but their mass pattern inherits the Z_3 structure because the Bergman kernel of D_{IV}^5 has N_c -fold periodicity in the angular coordinate.

Step 2: The Koide parameter Q measures the ratio of “average mass” to “average root-mass squared.” In spectral language: $Q = (\text{first moment}) / (\text{zeroth moment})^2$ on the Bergman spectrum. For a rank-2 domain with N_c angular sectors:

$$Q = \text{rank} / N_c = 2/3$$

This is a geometric identity: the rank counts independent spectral coordinates (radial), while N_c counts angular sectors. The ratio is the fraction of spectral weight in the radial direction.

Step 3: The Koide angle. The angular parameter θ_0 satisfies:

$$\cos(\theta_0) = -19/28$$

where $19 = n_C^2 - C_2 = 25 - 6$ and $28 = T_g = g(g+1)/2 = 7 \cdot 8/2$. Both are forced by the geometry.

Step 4: Precision. The BST prediction is $Q = 2/3$ EXACT. The observed value $Q_{\text{obs}} = 0.666658\dots$ deviates by 0.001%. This deviation is consistent with radiative corrections ($\alpha/\pi \sim 0.002$ at one loop). BST predicts the tree-level relation is exact; loop corrections dress it.

4. Predictions and Falsification (~0.5 pages)

1. Q remains $2/3$: Any future high-precision measurement of lepton masses will confirm $Q \rightarrow 2/3$ (not drifting away). The tau mass should converge toward the Koide-predicted value 1776.97 MeV.
2. Quark Koide: The same formula applies to quarks within each sector:
 - Up-type: $Q(u,c,t) = \text{rank}/N_c = 2/3$ (currently 0.34% from $2/3$)
 - Down-type: $Q(d,s,b) = \text{rank}/N_c = 2/3$ (currently ~3% from $2/3$, larger due to confinement dressing)
3. No fourth generation: Z_3 is exhaustive — exactly three generations, not four. If a fourth-generation lepton is found, BST is falsified.
4. Neutrino Koide: If neutrino masses satisfy Koide with $Q = 2/3$, the normal hierarchy is confirmed and $\text{sum}(m_\nu)$ is predicted.

5. Discussion (~0.5 pages)

- Why this works: $Q = 2/3$ is the ratio of two SMALLEST topological invariants of D_{IV}^5 ($\text{rank} = 2, N_c = 3$). Topological ratios cannot receive perturbative corrections. This explains the extraordinary precision.
- The Z_3 symmetry in Koide models is not ad hoc — it IS the color charge $N_c = 3$ of the Standard Model, appearing in the lepton sector through the unified geometry.
- Connection to the Standard Model: the same geometry that gives $Q = 2/3$ also gives $\alpha^{-1} = 137 = N_c^3 * n_C + \text{rank}$ and $m_p = 6\pi^5 m_e$.

Key Numerical Verification (from Toy 1734)

Quantity	BST	Observed	Precision
Koide Q	$2/3 = 0.66667$	0.66666	0.001%
$\cos(\theta_0)$	$-19/28 = -0.67857$	-0.67855	0.003%
$19 = n_C^2 - C_2$	$25 - 6 = 19$	—	exact
$28 = T_g$	$7*8/2 = 28$	—	exact
GM deviation	$\text{rank} * \Phi_3(C_2)/g = 86/7$	12.295	0.08%

References (key ones for PRL)

- Koide (1981), Phys. Lett. B 120, 161
- Foot (1994), Phys. Rev. D 49, 3617 (Z_3 model)
- Rivero & Gsponer (2005), arXiv:hep-ph/0505220

- Brannen (2006), arXiv:hep-ph/0606073 (Koide waterfall)
 - BST Working Paper (Koons, 2026), Zenodo DOI: 10.5281/zenodo.19454185
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Writing Assignments

- Lyra: Sections 2-3 (geometry + derivation). The mathematical meat.
 - Keeper: Sections 1, 4-5 (framing, predictions, discussion). Keep it accessible.
 - Casey: Abstract, final polish, submission logistics.
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Target: draft complete within 1 week. Submit to PRL by May 15, 2026.